

HIS/RIS Setup and Troubleshooting

[\(Click Here for Summary Version\)](#)

1 Personnel Requirements

Required Persons	Preliminary Reqs	Procedure	Finalization
1	Not Applicable	15 mins	Not Applicable

2 Overview

This procedure documents the set up and calibration of the system to a clinical hospital information system (HIS) or a radiology information system (RIS) .

3 Preliminary Requirements

3.1 Required Conditions

Condition	Reference	Effectivity
Acquire HIS/RIS, filming network information from the customer administrator.	-	-

4 Procedure

4.1 Setup

NOTE: All entry fields are case sensitive.

1. Collect and write down the following information about the HIS/RIS server:
- IP address

◦ Port number

◦ AE title

◦ Host name

You will need this information to complete this procedure.

2. From the Common Service Desktop, select the [Configure] button, select the [Guided Install] button, and in FE Mode login as *root* (password is *operator*).
- NOTE:** It is possible that the customer changed the default password. If you cannot log in, contact the customer for the correct password.
3. Under [Connectivity], select [HIS/RIS].

Illustration 1: HIS/RIS Screen

HIS/RIS

Hospital Name: GEHS

Suite ID: mr

Host ID: 32

Local AE Title (Hostname) (3-16 Char): t32

Local IP Address (Host IP): 3.7.25.32

Local Port Number (Host Port) (0-65535): 4000

Netmask: 0xffffc00

DICOM Network Timeout: Slow (300sec)

Default Router IP Address: 3.7.27.254

Station Name (1-8 Char): mr32

Buttons: [Configure] [Previous] [Next] [Home]

NOTE: All characters in the fields are case-sensitive. Be sure to type information accurately.

4. Enter applicable information in all fields (as provided by site) and click the [Configure] button.
5. Under [MWL Configuration], select [MWL Config Details].

Illustration 2: MWL Config Details Screen

MWL Config Details

Note: Changes will be effective only after application is restarted

Patient Detail Fields	Dicom Tag	Required (Y or N)
Name	(0x0010,0x0010)	✓
Patient ID	(0x0010,0x0020)	✓
Weight	(0x0010,0x1030)	✓
Date Of Birth	(0x0010,0x0030)	✓
Age	(0x0010,0x1010)	✓
Sex	(0x0010,0x0040)	✓
Scheduled Date	(0x0040,0x0007):(0x0040,0x0008)	✓

Exam Detail Fields	Dicom Tag	Required (Y or N)
Accession	(0x0008,0x0050)	✓
Exam Description	(0x0008,0x1030)	✓
SPS Description	(0x0040,0x0007):(0x0040,0x0008)	✓
Procedure ID	(0x0040,0x1001)	✓
Operator	(0x0008,0x1070)	✓
Radiologist	(0x0008,0x1060)	✓
Referring Physician	(0x0008,0x0090)	✓
Local Protocol		✓

Other Detail Fields	Dicom Tag	Required (Y or N)
Other Patient ID	(0x0010,0x1000)	
Current Location	(0x0038,0x0300)	
Special Needs	(0x0038,0x0050)	
Requested Contrast Agent	(0x0032,0x1070)	
Station	(0x0010,0x1110)	

Buttons: [Configure] [Previous] [Next] [Home]

Illustration 3: MWL Config Details Screen (Continued)

The screenshot shows the 'MWL Config Details' window. The left sidebar contains a tree view with the following items: Home, Basic Configuration (Network Info, System Configure, Hardware Configure, Set Time/Date), Connectivity (HIS/RIS), MWL Configuration (MWL Config Detail, Return Tag Config, SCP Configure, Table Column Sele, PPS Configure), PM Configuration (Auto Mapping Con, Password Config, Default Configur), Preferences (Edit Patient Data, Set Exam Number, Set Disk Limit, HIPAA, System Preferences, SAR ddt Level, Autovoice, Patient Anonymiza), Utilities (Save/Restore, Display MOD/CD-Patches, DBReset Image/Er, Sprsnap, Log file, Option Key), and a bottom section for 'Enter for CTR Device'. The main area is titled 'MWL Config Details' and contains a table of fields with checkboxes for selection.

Field	Dicom Tag	Required (Y or N)
Exam Description	(0x0008,0x1030)	☒
SPS Description	(0x0040,0x0007):(0x0040,0x0008)	☒
Procedure ID	(0x0040,0x1001)	☒
Operator	(0x0008,0x1070)	☒
Radiologist	(0x0008,0x1060)	☒
Referring Physician	(0x0008,0x0090)	☒
Local Protocol		☒

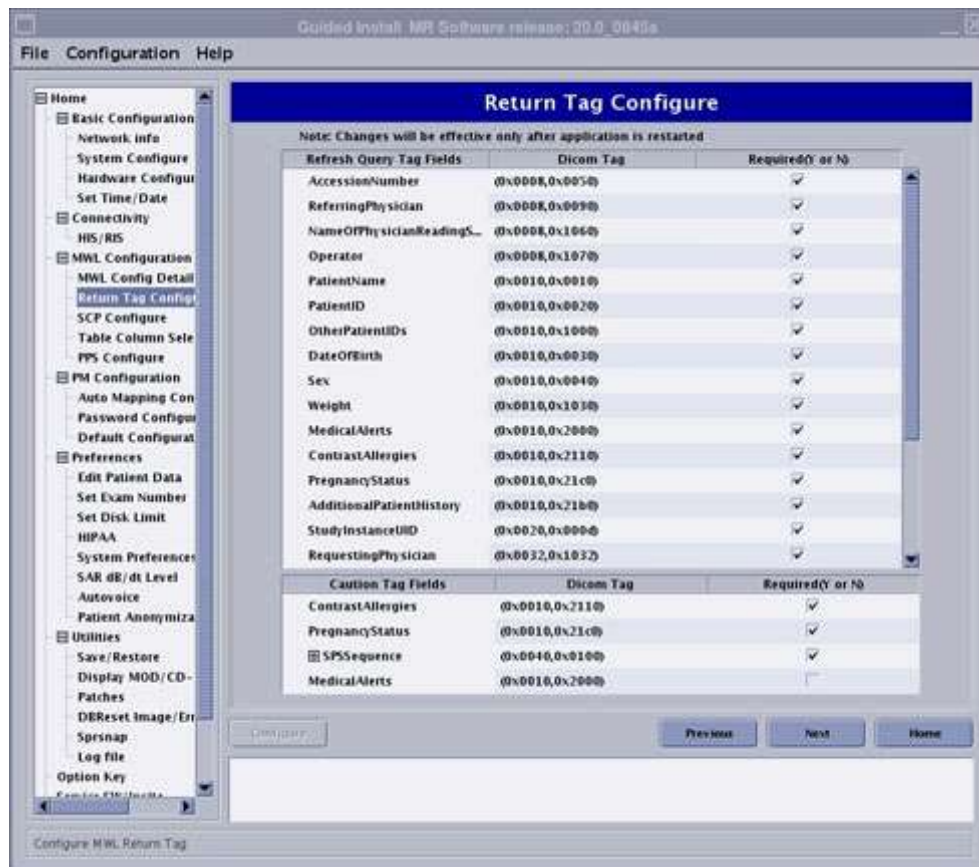
Other Detail Fields	Dicom Tag	Required (Y or N)
Other Patient ID	(0x0010,0x1000)	☐
Current Location	(0x0038,0x0300)	☐
Special Needs	(0x0038,0x0050)	☐
Requested Contrast Agent	(0x0032,0x1070)	☐
Allergies	(0x0010,0x2110)	☒
Pre-Med	(0x0040,0x0012):(0x0040,0x0013)	☒
Pregnancy Status	(0x0010,0x21c0)	☒
History	(0x0010,0x21b0)	☒

Search Detail Fields	Dicom Tag	Required (Y or N)
Name	(0x0010,0x0010)	☒
Patient ID	(0x0010,0x0020)	☒
Accession	(0x0008,0x0050)	☒
Referring Physician	(0x0008,0x0090)	☐
Procedure ID	(0x0040,0x1001)	☒

At the bottom of the window, there are buttons for 'Configure', 'Previous', 'Next', and 'Home'. The status bar at the bottom left reads 'MWL Config Details (panel)'.

6. If any of fields, other than the defaults, are desired to appear in the Modality Worklist, select or deselect appropriately and click the [Configure] button.
7. Under [MWL Configuration], select [Return Tag Configure].

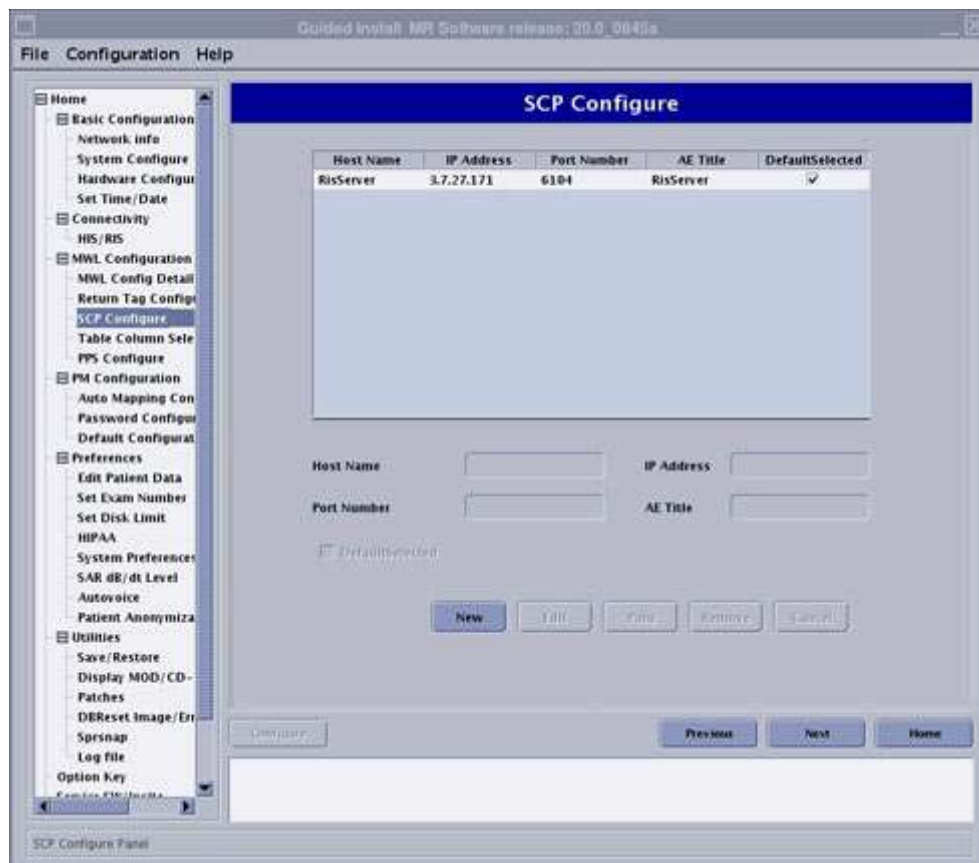
Illustration 4: Return Tag Configuration Screen



8. If any fields, other than the defaults, are desired by the site to be refreshed in the Modality Worklist, select or deselect appropriately and click the [Configure] button.

9. Under [MWL Configuration], select [SCP Configure].

Illustration 5: SCP Configure Screen



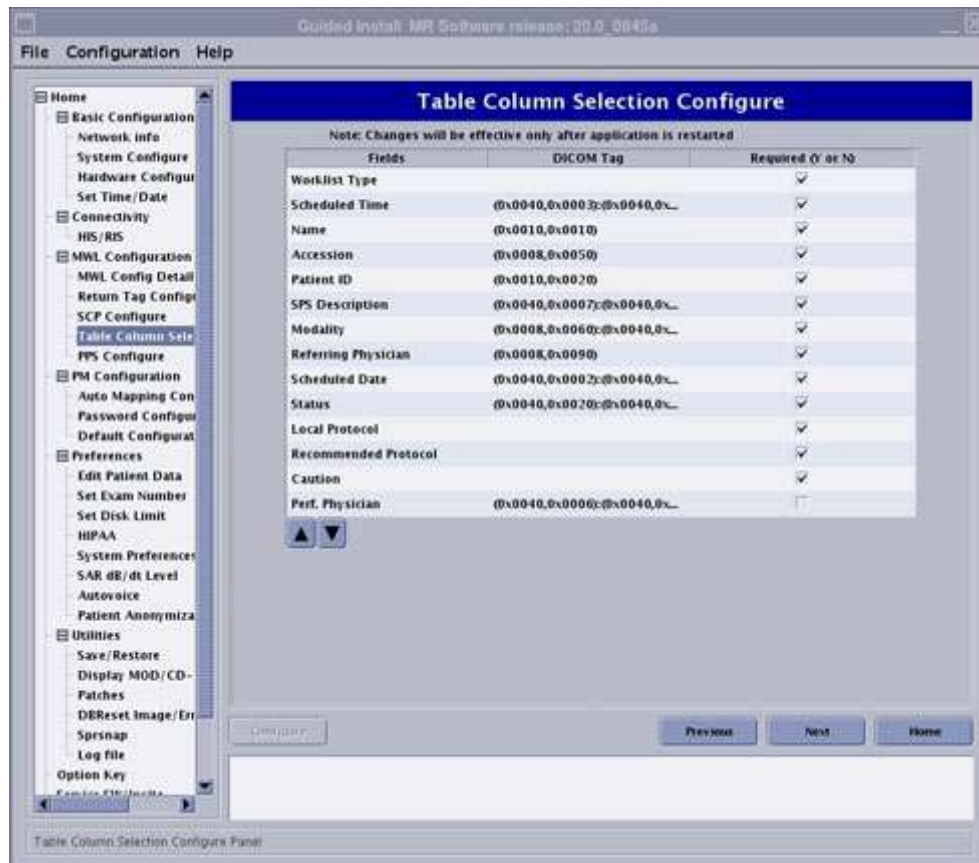
10. Click the [New] button to set up a new item.
11. Enter applicable information in all fields (as provided by site) and click the [Configure] button.
12. To verify network connectivity, click the [Ping] button.

NOTE: The Ping command may take up to three minutes to complete.

If the Ping command fails, start a command window and ping the IP address. Also, you can try to telnet to the IP address from the command window.

13. Under [MWL Configuration], select [Table Column Selection Configure].

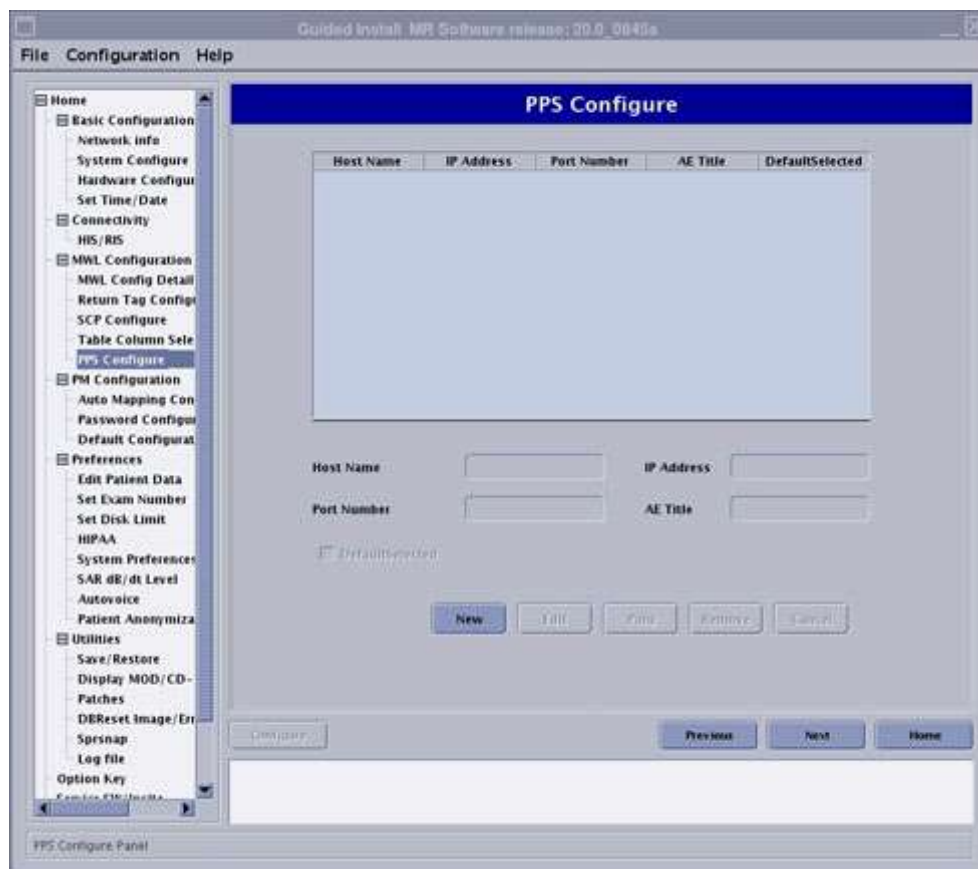
Illustration 6: Table Column Selection Configuration Screen



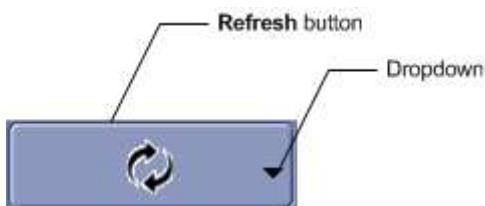
14. Select or deselect any fields desired to be displayed in the patient record table (located at the top of the scan window) and click the [Configure] button.
15. Under [MWL Configuration], select [PPS Configure].

NOTE: At release of this document, PPS Configure is available with 1.5T systems only. PPS Configure will be available for 3.0T at a later time.

Illustration 7: PPS Configure Screen



16. Click the [New] button to set up a new setup.
17. Enter applicable information in all fields (as provided by site) and click the [Configure] button.
18. Go into the scan desktop and click the drop down on the [Refresh] button to display the RefreshData dialog. This dialog allows you to set up the default query run when the Refresh button is clicked. Make the changes requested by the site and then click [Save].



RefreshData

Breadth of Search

With Date Range

☒ Today

☐ Any Day

☐ Days before Today 0

☐ Days after Today 0

☐ Date Range

MM-DD-YYYY

From - -

MM-DD-YYYY

To - -

Search RIS for:

☒ This System

☐ This Modality

Enter AE Title (Optional)

☐ All Modalities

Refresh settings

☒ Refresh Worklist on Startup

☐ Refresh worklist only when refresh button is clicked

☒ Auto-RefreshTiming 0 Minutes

4.2 Functional Check

1. Click the [Work List] icon for the scan screen.

Illustration 8: Work List Icon



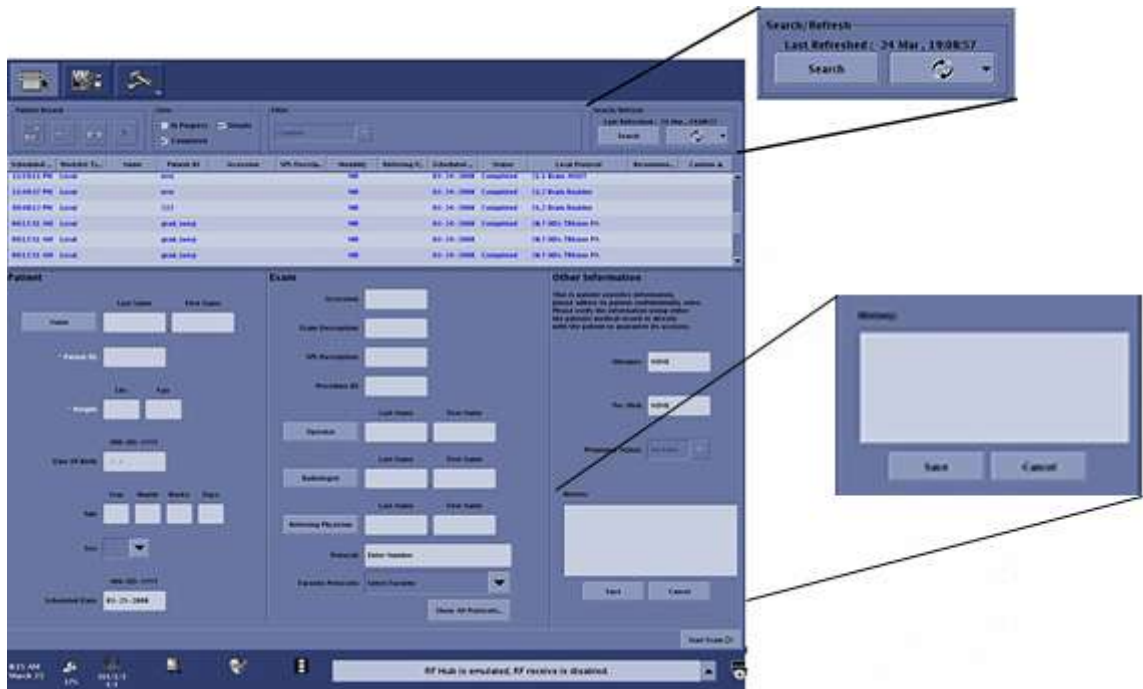
2. To confirm the connectivity of the system to the RIS system, go into the scan desktop and select the [New Patient] icon.

Illustration 9: New Patient Icon



3. On the scan screen enter the following:
 - a. **Patient ID:** geservice
 - b. **Weight:** 111
4. Click the [Save] button.
5. In the Search/Refresh box, click the [Search Data] button.

Illustration 10: Search Button



The Search Data window displays.

Illustration 11: Search Data Window

Search Data

Name

Patient ID:

Accession:

Procedure ID:

Breadth of Search

With Date Range

☒ Today
 ☐ Any Day
 ☐ Days Before today
 ☐ Days After Today
 ☐ Date Range

0

0

MM-DD-YYYY

From

MM-DD-YYYY

To

Search RIS for:

☒ This System
 ☐ This Modality
 ☐ All Modalities

Enter AE Title (Optional)

Clear

Search

Close

From this page select how you want to pull information from the RIS system. The Get Patient List options are:

1. Enter a patient Name, Patient ID, Accession Number, or Procedure ID to define the search.
2. Selecting This System, which pulls the patient lists entered for this system.

file:///C:/Users/MICHA/OneDrive/Work Docs/GE/Manuals/1.5T HDxt 23x/root/data/Signal_EXCITE/content/1137178.htm

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3. Selecting **This Modality** pulls the patient list for all MR systems at this facility.
4. Selecting **All Modalities** pulls the patient list for all systems associated with this facility. Using this option may cause the scanner to hang if the facility has too large of Work List for all scanners at the site.
5. Selecting **Date Range** pulls patient lists for any set day or date range.
6. After entering the proper Search Data, click the [Search] button.

Illustration 12: Search Details Selected

Search Data

Name:

Patient ID:

Accession:

Procedure ID:

Breadth of Search

With Date Range

☒ Today

☐ Any Day

☐ Days Before today:

☐ Days After Today:

☐ Date Range

MM-DD-YYYY

From:

MM-DD-YYYY

To:

Search RIS for:

☒ This System

☐ This Modality:

Enter AE Title (Optional)

☐ All Modalities

Clear Search Close

7. The results appear in the list above the patient information.

Illustration 13: Full Work List

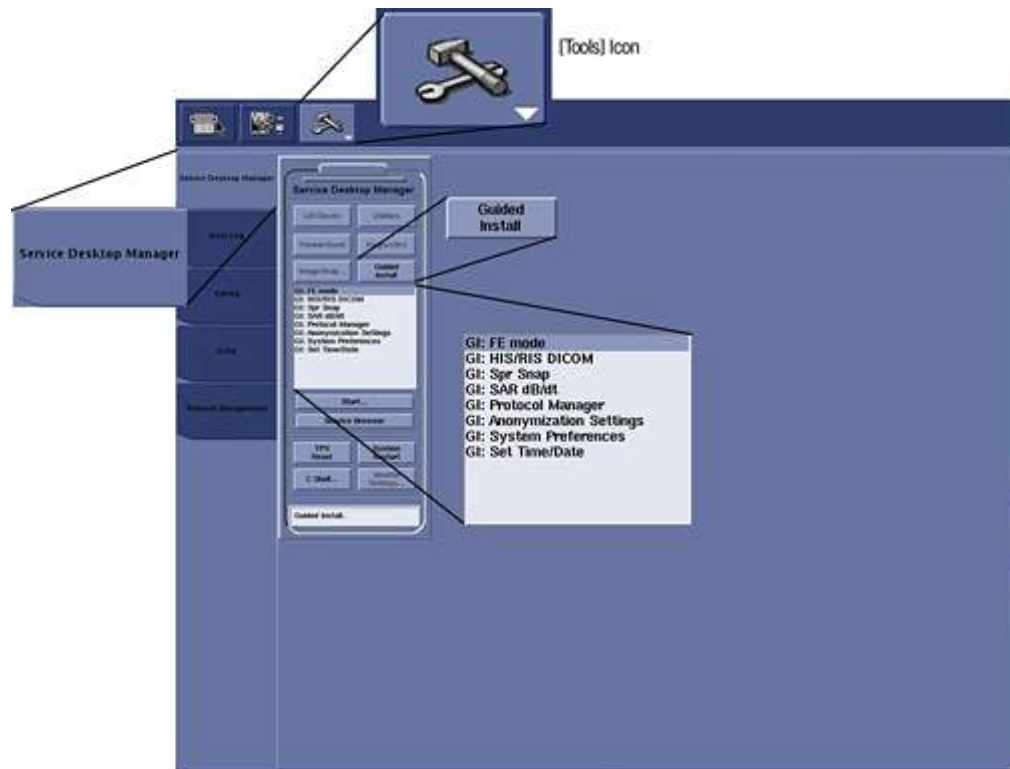
Scheduled	Modality	Name	Patient ID	Accession	SPS Device	Modality	Referring CL	Scheduled	Status	Local Protocol	Accession	Cancel
08/19/17 PM	Local	MRECHO	1001			MR	01-12-2008	Completed	24.8 MR-0.0a			
08/19/17 PM	Local	MRECHO	1002			MR	01-12-2008	Completed	24.8 MR-0.0a			
08/19/17 PM	Local	MRECHO	1003			MR	01-12-2008	Completed	24.8 MR-0.0a			
08/19/17 PM	Local	MRECHO	1004			MR	01-12-2008	Completed	24.8 MR-0.0a			
08/19/17 PM	Local	MRECHO	1005			MR	01-12-2008	Completed	24.8 MR-0.0a			
08/19/17 PM	Local	MRECHO	1006			MR	01-12-2008	Completed	24.8 MR-0.0a			
08/19/17 PM	Local	MRECHO	1007			MR	01-12-2008	Completed	24.8 MR-0.0a			

Search Results

4.3 Troubleshooting

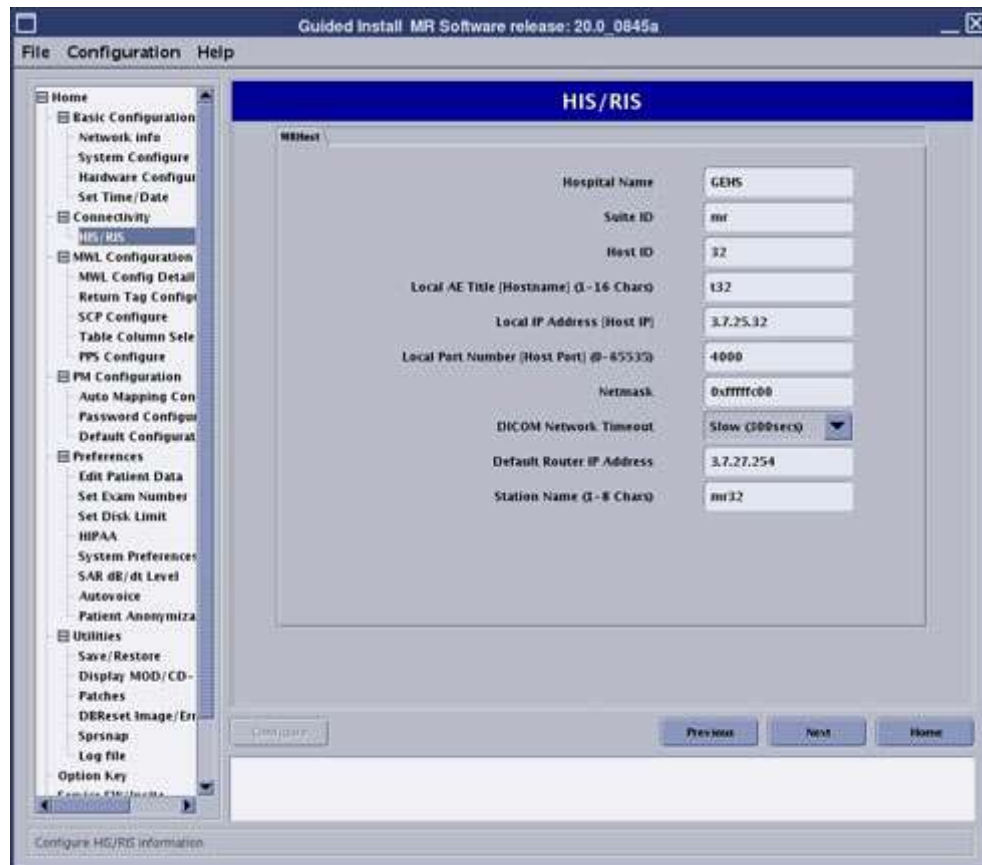
1. From the Common Service Desktop, select the [Guided Install] button, and in FE Mode login.

Illustration 14: Accessing HIS/RIS Setup



2. Under [Connectivity] select [HIS/RIS].

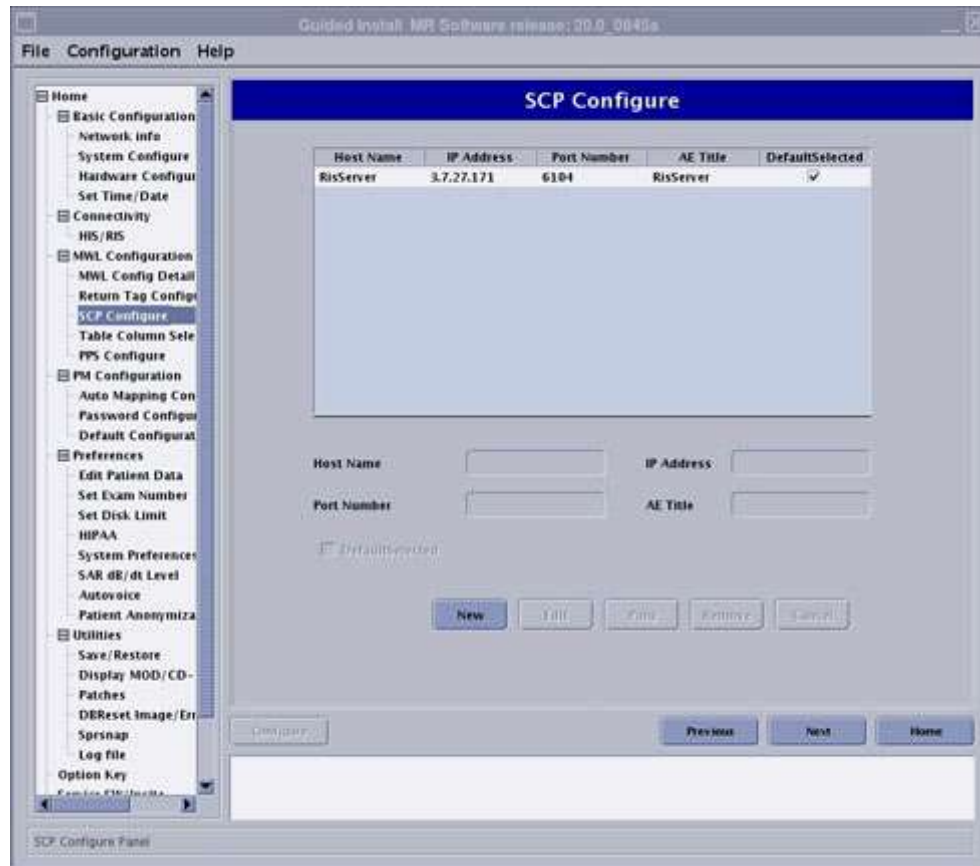
Illustration 15: HIS/RIS Setup Screen and Tabs



3. Ensure that the **Hospital Name**, **Suite ID**, **Host ID**, and **Local AE Title** are correct for the MR system. The **Local Port Number (Host Port)** should be set to **4000** (same as the DICOM (Digital Imaging and Communications in Medicine) networking port number for MR systems). These are the parameters that the hospital RIS system uses to identify our system.

4. Under [MWL Configuration], select [SCP Configuration].

Illustration 16: SCP Configure Screen



Ensure the following information with the site network administrator:

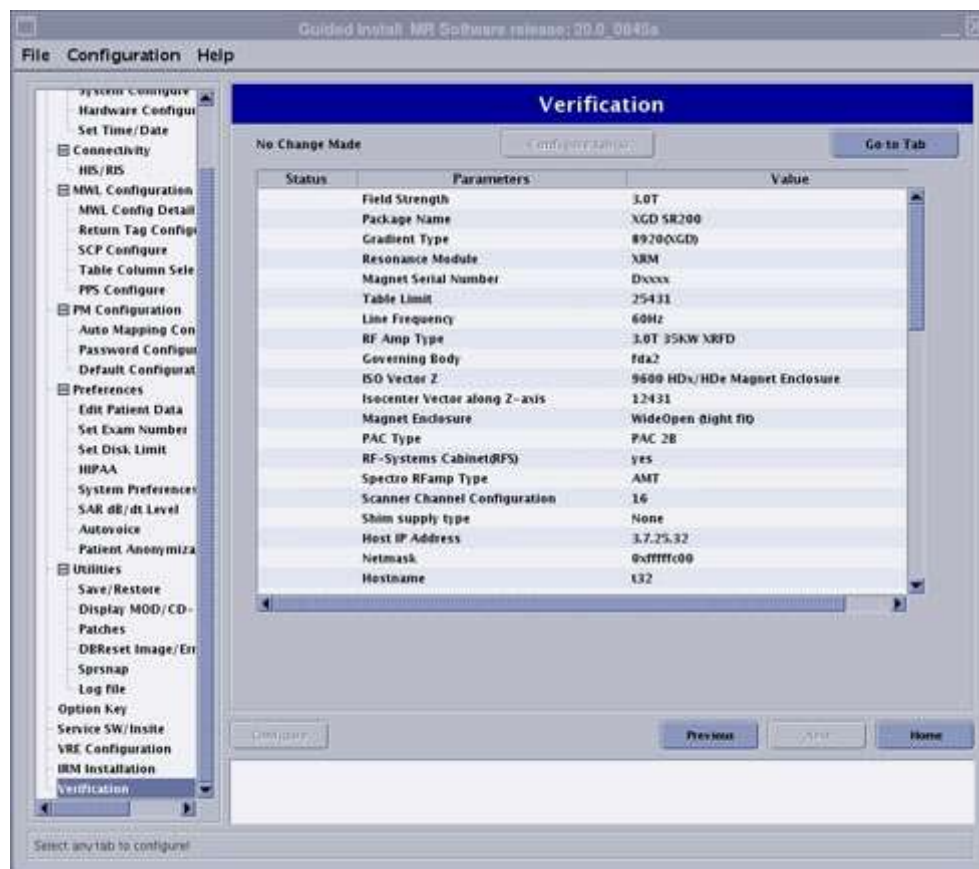
- **Host Name** is the site name.
- **AE Title** is the Modality Work List (MWL) Application Entity (AE) title of the RIS system of the site.
- **IP Address** is the internet protocol (IP) address of the RIS server.
- **Port Number** is the port number of the RIS system of the site.

NOTE: The Ping command may take up to three minutes to complete.

If the Ping command fails, start a command window and ping the IP address. Also, you can try to telnet to the IP address from the command window.

5. Under [Home], select [Verification].

Illustration 17: Verification Screen



6. Select [Configure] and confirm no errors.

7. Select File > Quit.

5 Finalization

No finalization steps.

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